

Aim :-

Write Program to implement flow control at data link layer using SLIDING WINDOW Protocol Simulate the flow of frames from node to node.

create a Sender Program:

1. Input window size from the user.
2. Input a Text message from the user.
3. Consider 1 character per frame.
4. Create a frame with following field [Frame no, DATA].
5. Send the frame [Prior the Output on Server and save it in a file called Sender-Buffer].
6. Wait for the acknowledgment from the receiver [Induce delay in the program].
7. Read a file called Receiver-Buffer.
8. Check ACK field the Acknowledgement number.
9. If the Acknowledgement number is as expected, send new set of frames according. [overwrite the Sender-Buffer file with new frames]. Else it is not expected, resend.

create a Receiver file with following feature.

1. Read a file called Sender-Buffer.
2. Check the Frame no.
3. If the Frame no are as expected, write the appropriate ack no in the Receiver Buffer file. Else write it Back is received Buffer file.

Program:

```
from re import *
import time
import os
os.system("clear")
SB = open('Sender-Buffer.txt', 'wt')
RB = open('Receiver-Buffer.txt', 'wt')
SB.truncate(0)
RB.truncate(0)
ws = int(s)
if (ws < len(s)):
    for i in range(10, len(s), ws):
        P = s[i : i + ws]
        Y = s[i + ws : ws + ws]
        Print("Sent -> " + str(P))
        time.sleep(ws)
        Print("Summary -> " + str(Y))
        x = 0
        while (x < ws):
            time.sleep(2)
            if (len(P) == x):
```

Print("ack - 1, P[x] "1.")

RB.write(P[x])

time.sleep(1)

if (len(y) > k):

Print("Sending -> ", y[x])

SB.write(y[x])

x += 1

else:

Print("The window size is too large")

Result:

Thus the implementation of the flow control at data link layer is executed successfully.